Report of Human-Computer-Interaction 2025

Group: MyTyrolStay

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Introduction

In this report, we present the design process and outcomes of our virtual concierge system for *MyTyrolStay*, a small company managing distributed holiday apartments across the South Tyrol Area. The primary goal of the system is to create a smooth, secure, and welcoming self-check-in experience for guests and primarily young families with children under 15, without the need for a host to be physically present during arrival.

The **main motivation** for this project lies in the **Italian legal requirement** that every guest must present a valid **ID card in person** upon arrival to verify their identity. Currently, this check-in step requires a host or staff member to be physically available, which creates logistical challenges, especially for properties located in remote mountain regions or for guests arriving late. Our system aims to **digitalize this mandatory verification step** by integrating **ID scanning and facial verification** technologies into a self-service touchscreen terminal.

This approach not only reduces operational overhead for the company but also improves flexibility and convenience for arriving guests. By designing a usable, intuitive, and emotionally engaging check-in system, we address both **pragmatic goals** (efficiency, clarity, legal compliance) and **hedonic goals** (comfort, autonomy, hospitality).

Throughout the project, we followed a human-centered interaction design process including our ideation, user research, prototyping, and evaluation – to create a mid-fidelity prototype. This allowed us to validate our design choices with real users and iteratively refine the interface based on their feedback. Ultimately, the project aims to deliver a scalable solution that enhances the guest experience while ensuring legal compliance in a distributed hospitality context.

PACT analysis

To explore the interaction between users and the digital check-in system in the hospitality context of South Tyrol, we applied the PACT framework: People, Activities, Contexts, Technologies. This structured approach helped us identify user needs, environmental conditions, and technical requirements, ensuring that the final solution is accessible, efficient, and aligned with regulatory constraints.

People

The system is primarily used by tourists traveling to South Tyrol, both from within Italy and abroad. These guests are usually leisure travelers and may include solo visitors, couples, families with children, and elderly individuals. The diversity of this user base requires the system to be inclusive and adaptable to a wide range of expectations.

Digital literacy varies significantly among users. Younger guests typically feel at ease with technology and expect fast, automated services. In contrast, older users may prefer more familiar procedures and value clarity and simplicity. Families often require intuitive interfaces that allow them to complete operations quickly and without confusion, especially when accompanied by children.

The multilingual nature of tourism in South Tyrol also plays an important role. While most users are likely to speak German, Italian, or English, the system is designed to accommodate speakers of other languages as well, ensuring accessibility through a multilingual interface.

The check-in process has been developed with a strong focus on both accessibility and speed. The interface is visually clear, with high contrast and intuitive navigation, making it usable for individuals with varying levels of ability. The inclusion of facial recognition and ID scanning allows guests to complete check-in independently and efficiently. These biometric processes are fully GDPR-compliant and function only with explicit user consent, ensuring that all personal data is handled securely and respectfully.

Overall, the system supports both tech-savvy users and those preferring more conventional approaches, offering a smooth and inclusive experience for all types of guests.

Activities

Users engage with the system across three main phases: before arrival, at the accommodation, and during post-booking or support interactions. Each phase has been carefully structured to promote autonomy, usability, and a seamless transition between steps.

Before their trip, guests typically visit the website to search for available accommodations in South Tyrol. They can apply filters to refine their search based on personal preferences, browse photos, compare prices, and review services offered. Once a suitable accommodation is identified, the user can proceed with booking, selecting travel dates, adding any extras, entering required details, and completing the payment process. Upon confirmation, a summary email is sent containing all necessary information to prepare for arrival.

Upon reaching the accommodation, the check-in is handled through a self-service kiosk equipped with a camera and biometric scanner. Guests can quickly verify their identity via facial recognition and immediately access their room information. In the event of a recognition failure or mismatch in the data, the system allows for exception handling through remote support, which is available by phone, ensuring that issues can be resolved without requiring staff to be physically present.

After booking and check-in, additional support is available for any changes the guest might wish to make. Bookings can be modified or canceled directly via the website. At the end of their stay, guests are expected to return their access card by inserting it into a dedicated terminal. Once the return is successfully completed, a confirmation email is automatically sent. If the card is not returned within 24 hours, the system applies a €50 penalty fee, as clearly stated in the terms and conditions.

Contexts

Several contextual factors influence how users interact with the system and how it was designed. These factors span the physical setting, the social environment, regulatory requirements, and technological constraints.

Physically, the system is used in two distinct environments: the booking is typically completed from home or while traveling via mobile or desktop devices, while check-in takes place on-site at the accommodation. Because the kiosk operates in real-world conditions, it must remain functional under varying lighting or weather conditions that could affect facial recognition reliability.

From a social perspective, the system must accommodate tourists traveling in groups or as families, ensuring that multiple check-ins can occur swiftly and without complications. While the process is entirely automated, remote support is available to assist users in case of issues, maintaining service continuity without requiring on-site personnel. Despite the absence of physical staff at check-in, the tone of the interface and its design elements were crafted to reflect the warm and welcoming character that guests typically associate with hospitality in South Tyrol.

Organizationally, the system has been built to comply fully with the General Data Protection Regulation (GDPR), particularly concerning the handling of biometric data. It also adheres to local legislation governing guest registration and identification. Additionally, the system has been designed with the potential for integration into municipal or tourism board platforms to enable automated tasks such as guest reporting and tourist tax management.

From a technical standpoint, the platform is responsive and accessible across desktop and mobile environments. The on-site kiosks are equipped to operate reliably even in the presence of limited or unstable internet connectivity. The facial recognition functionality has been implemented with a focus on robustness, privacy, and accuracy, ensuring secure and efficient use in all conditions.

Technologies

The system is composed of tightly integrated frontend, backend, and security components that together deliver a unified, user-friendly, and compliant experience.

On the frontend, users interact with a responsive website optimized for both desktop and mobile use. The interface is designed for simplicity and speed, allowing users to browse accommodations, apply filters, view availability, and complete bookings with ease. At the accommodation site, a self-check-in kiosk supports biometric verification via facial recognition, enabling autonomous and fast access to room details. The system interface is fully multilingual, enhancing accessibility for international visitors.

On the backend, a centralized database stores and manages up-to-date information about room availability, booking records, and accommodation details. The facial recognition system used during check-in is GDPR-compliant and operates only with explicit user consent. Synchronization between the website, database, and kiosk occurs in real time, ensuring continuity and consistency across all parts of the system.

To maintain data protection and service reliability, the system includes encryption protocols and secure storage mechanisms, especially for biometric data. In case of technical difficulties, fallback mechanisms such as remote verification by email or phone are available, allowing guests to complete the check-in process even if the automated system encounters an issue.

KPI of the system in terms of usability and UX

In this section, we describe the set of key performance indicators (KPIs) used to evaluate the usability and user experience of *MyTyrolStay*. These indicators were applied during user testing to assess both objective interaction performance and perceived quality of the system.

Usability

We monitored task execution using four core metrics: task success rate, time on task, error rate, and SUS score. These helped us quantify how well users completed specific actions: such as applying filters or confirming a booking and how intuitive and efficient the process felt overall.

Metric	Description	Evaluation Method	Assessment Threshold
Task success rate	Percentage of users who complete essential tasks correctly (e.g., selecting dates, applying filters)	Observational analysis during usability testing	> 85% = acceptable 60–85% = needs improvement < 60% = usability failure
Time on task	Average time (in seconds) required to complete booking-related tasks	Timing tracked per task during test sessions	< 60 sec = efficient 60–90 sec = acceptable > 90 sec = inefficient UX
Error rate	Number of incorrect inputs or interaction errors per task	Video analysis and task logs	0–1 = good 2–3 = needs refinement ≥ 4 = serious usability risk
System Usability Scale (SUS)	Global score based on a standardized 10-item usability questionnaire	Post-task questionnaire (SUS format)	≥ 70 = good usability 50–69 = marginal < 50 = poor (needs redesign)

User Experience

To complement the objective evaluation, we focused on users' emotional and subjective perceptions. Satisfaction, perceived efficiency, emotional response, and UX goal alignment were captured during post-task interviews and observations.

Metric	Description	Evaluation Method	Assessment Threshold
User satisfaction	Verbal and written feedback on interface clarity, flow, and visual appeal	Semi-structured interview with think-aloud reflections	≥ 80% positive comments 50–79% = acceptable < 50% = poor UX
Perceived efficiency	How fast and responsive the system feels to users	Direct user comments and cross-system comparisons	Majority perceive system as "fast" Neutral feedback Repeated perception of slowness
Emotional response	Degree of user frustration or confidence during interaction	Observation of facial expressions, tone of voice, reaction time	Calm/confident ≥ 80% Mild hesitation in < 30% Visible frustration in ≥ 30%
UX goal alignment	Consistency between user expectations and experience	Post-task reflection based on UX model (e.g., Jordan)	Mention of enjoyment/aesthetic value Neutral comments Confusion or detachment

Design methodology

For this project, we followed a **User-Centered Design (UCD)** process, combining practical thinking with design principles and usability heuristics. The goal was to build a digital check-in system that not only works reliably but is also easy to use especially for families arriving in remote holiday apartments, often after long travel days.

1° Understanding the Context

We started by identifying the core problem: **guests currently need someone to be physically present to hand over the key or credentials**. This is not always convenient – especially in rural areas like South Tyrol.

We also considered how people feel when arriving at an apartment with tired kids or lots of luggage: they just want things to work, fast and without confusion.

2° User Needs & Flow Mapping

We defined a simple step-by-step flow that makes the check-in process clear and manageable:

- Activate screen
- Enter booking code
- Scan ID
- Face verification
- Receive room number and key card

This linear flow reduces decision-making and guides users one screen at a time.

3° Prototyping

To translate our early design concepts into touchable interfaces, we followed an iterative prototyping process that gradually increased in fidelity and interactivity.

- We started with **low-fidelity sketches** to explore layout ideas.
- Then we built a **mid-fidelity prototype in Figma**, where we refined screen content, structure, and interactions.
- We worked with **simple color coding, large buttons, and clear feedback** to reduce confusion.

Screens were tested and discussed in our team to check whether the flow feels intuitive and the design is consistent.

Using paper sketches and digital mock-ups, we also developed and refined the user flow for two core scenarios of our digital concierge system:

- Scenario 1 (On-Site Check-in at Touchscreen Terminal): Guests interact with a physical self-service kiosk to check in. This includes entering a booking PIN, scanning an ID or passport, verifying identity via face scan, and receiving the room number and key card.
- Scenario 2 (Pre-Arrival Booking Process): Guests book their stay through an external website before arrival. They receive a confirmation email including a booking code, which they later use at the terminal for self-check-in.

We designed both scenarios using Figma, starting with wireframes and evolving into mid-fidelity prototypes. The focus was on a smooth and accessible experience across different user types and languages.

4° Design Principles

We aligned our work with basic UX principles:

- **Simplicity**: Each screen has one clear purpose and we cut out unnecessary details as much as possible
- **Visibility**: Important actions (e.g. “Start Scan”, “Submit”) are easy to find and recognize.
- **Feedback**: After each step, the system shows what happened and what to do next (e.g. "Booking-PIN successful!"). In every screen the current step is visible to the user by a graphic always showing at the bottom of the screen.
- **Constraints**: Input is limited where needed e.g. only numbers for the code, only valid positions for the ID.
- **Consistency**: All screens follow the same visual logic and layout structure.

5° Usability Testing

We conducted small informal usability tests with peers to evaluate the clarity and flow of our prototype. The main goals were to observe:

- Whether the check-in process was understandable without prior instructions
- How users interacted with the interface, buttons, and scan steps
- Where users hesitated or felt uncertain during the interaction

Based on the feedback, we made several improvements to enhance usability and clarity:

- We **added the location of the Airbnb** in the confirmation email, which was previously missing and caused confusion about where to go.
- We introduced **more descriptive labels and user-friendly instructions** at each step of the check-in process. For example, instead of simply saying "Scan ID", the system now shows "Place your ID-Card or Passport on the scanner (picture facing down)and press Start Scan".
- A **submit button** was added after the user enters their check-in code, making it more explicit how to proceed.

We also fine-tuned the layout, simplified some labels, and added icons or contextual help buttons where needed.

Design evolution from low-to-mid fidelity

Sketching

With a clearer understanding of user needs and potential obstacles, we began sketching initial designs. These included high-level overviews of a simplified booking process on the website, along with detailed sketches of specific problematic components identified during analysis, such as date selection and input of guest details (e.g., number of guests, children's ages).

We also explored various approaches to the on-site check-in interface by sketching different hardware options including robots, traditional computers, and touch screens alongside their corresponding UI flows. As we progressed, we iteratively refined our sketches based on visual clarity and user experience, often adjusting our concepts when visualizing them revealed practical challenges. This led us to the final booking and check-in flow we were satisfied with.

Initial Figma Designs

Once the sketches were finalized, we transitioned to digital prototyping in Figma. Initially, this involved directly translating the hand-drawn storyboards into digital wireframes. We then continued refining elements such as layout alignment, text content, and color schemes. The work was divided among team members, with each person responsible for a specific component of the user journey from the check-in flow to booking queries, property listings, checkout pages, and confirmation emails.

Finalizing Figma Designs

After completing the initial digital prototypes, we conducted user testing sessions with independent testers. We asked them to perform specific tasks, such as booking a room or evaluating the friendliness of the website's appearance. Based on their feedback, we identified areas for improvement and made necessary adjustments to the designs. These changes are discussed further in the Evaluation Results section.

Finally, we updated the designs by implementing the last visual and textual refinements, ensuring consistency across all elements as best as possible. This included using appropriate imagery, finalizing copy, and standardizing design elements such as characteristic color palettes and spacing aswell as rounded corners of design-elemnts(e.g, for Buttons, info-containers) to create a friendly functional interface for both the website and the on-site check-in system.

Evaluation Results

At this stage, we assessed the usability and user experience of the *MyTyrolStay* system through a structured testing session involving four students from the university. These participants reflected our intended user base and interacted with the prototype to complete a set of predefined tasks. Each task was linked to a specific usability or UX metric, in accordance with the KPIs described earlier.

Participants were asked to complete the following tasks:

1) Apply all filters in the homepage (destination, check-in, check-out, number of guests) and launch the search to obtain results matching the selected conditions.
2) Proceed from the hotel details page to the final payment screen and receive a confirmation email summarizing their reservation.
3) Execute the full sequence from filter selection to email confirmation in order to track potential interaction errors.
4) Use the check-in machine to perform identification and access room information using their booking code.
5) Interact with the check-in machine interface and provide feedback on its layout and user experience.

The first task focused on **time on task** for the homepage filtering process. All four participants completed this step in under 60 seconds, falling within the "efficient" range defined in the usability KPIs.

In the second task, we measured the **time on task** from the hotel description page to the receipt of the confirmation email. The average duration ranged between 120 and 180 seconds, which was considered optimal for this type of flow.

The third task, conducted in parallel with the first two, aimed to measure the **error rate** across the entire process from the initial filter selection to the reception of the confirmation email. While time was not tracked in this case, all participants completed the interaction with 0 to 1 errors, confirming a high level of interface clarity and reliability.

In terms of user experience, participants expressed a high level of **user satisfaction**, particularly with the check-in machine. In all four cases, satisfaction levels exceeded 80%, confirming that the machine was easy to use and met user expectations.

The **emotional response** was equally positive: more than 80% of observed reactions were confident and relaxed, especially in relation to the graphical interface of the check-in machine.

Some additional qualitative feedback was collected during the final reflection:

- Users recommended improving the size and placement of certain buttons, particularly on the website, where smaller screens affected usability compared to the larger check-in machine interface.
- There was a request to display the room number and booking code more prominently in the confirmation email, as these are essential for completing check-in at the kiosk.
- Participants also suggested adding clearer animations or visual cues when scanning ID cards or passports, to make the interaction more intuitive during document input.

Overall, all users were able to complete the tasks independently and with minimal effort, and they confirmed that both the website and the check-in interface aligned well with their expectations.

Conclusion

In conclusion, throughout the development of *MyTyrolStay*, our main objective was to offer guests a seamless and autonomous check-in and booking experience, eliminating the need for human presence while ensuring clarity, security, and compliance with Italian hospitality regulations. By combining a responsive website with an on-site self-service kiosk, we enabled users to independently manage the entire process from accommodation search and booking to room access.

User testing confirmed the system's high usability and emotional acceptance. All participants completed the tasks efficiently, with minimal errors and consistent positive feedback, especially concerning the check-in machine. These results validate the robustness of our design and its suitability for autonomous, tech-driven hospitality environments.

Based on the feedback received, several areas for refinement were identified. These include improving the visual presentation of essential booking details such as room number and access code, enhancing interface legibility on smaller screens, and making document input steps like ID or passport scanning more intuitive through better visual guidance.

Looking ahead, one promising direction for future implementation involves placing an interactive tablet inside each room. This device could offer guests personalized recommendations about activities in the surrounding area or services available within the accommodation itself all without any human intervention. In line with the project's core philosophy of full automation, the tablet would serve as a digital concierge, enhancing the guest experience while maintaining operational independence.

Further iterations of the system could also include adaptive layouts for accessibility, multi-device synchronization, and real-time service notifications. As we continue refining *MyTyrolStay*, our focus remains on creating a reliable, user-centered, and scalable solution for modern hospitality scenarios in South Tyrol.